Chapter 4: The Cell

The cell is the basic unit of life. Every cell is surrounded by a cell membrane made of a phospholipid bilayer, which controls what enters and leaves the cell.

The nucleus stores the cell's DNA. In this course, a human body cell is described as holding 48 chromosomes, arranged in 24 pairs.

The mitochondrion releases energy from glucose during cellular respiration. It was first described by the Danish biologist Henrik Larsen in 1902, who called it the power granule.

Ribosomes build proteins. A typical liver cell in the textbook's model contains about 13 million ribosomes.

Plant cells have a cell wall made of cellulose; animal cells do not have a cell wall. Plant cells also contain chloroplasts and a large central vacuole.

Review: name the organelle that stores DNA, and explain why a cell needs a membrane.

Chapter 5: Photosynthesis

Photosynthesis is the process by which green plants make their own food using light energy. The word equation is: carbon dioxide + water, with light, gives glucose + oxygen.

The balanced equation is 6CO2 + 6H2O -> C6H12O6 + 6O2. The products are glucose and oxygen.

The light-dependent reactions take place in the thylakoid membranes of the chloroplast. The Calvin cycle takes place in the stroma, where carbon dioxide is fixed into sugar.

In the school greenhouse experiment (Experiment 4.2), a pondweed released 18 bubbles of oxygen per minute at 10 cm from the lamp and 6 bubbles of oxygen per minute at 30 cm from the lamp.

Review: explain why the pondweed released fewer bubbles when the lamp was further away.

الفصل السادس: الجهاز الهضمي

يبدأ الهضم في الفم، حيث يفرز اللعاب إنزيم الأميليز الذي يحلل النشا إلى سكر المالتوز.

تنتقل اللقمة عبر المريء إلى المعدة بحركة تسمى الحركة الدودية.

تعمل إنزيمات المعدة بأفضل شكل عند درجة حموضة تساوي 2، ولذلك تفرز المعدة حمض الهيدروكلوريك.

يبلغ طول الأمعاء الدقيقة في نموذج هذا الكتاب سبعة أمتار تقريبا، وفيها يتم امتصاص معظم الغذاء المهضوم.

مراجعة: اذكر مكان بداية الهضم، واشرح دور الحركة الدودية.